Chapter 31

Capstone Multi-Omics Clinical Case Studies

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Abstract

Proteomics and metabolomics represent distinct biochemical disciplines that converge upon a unified computational grammar in R and Bioconductor. At the raw instrument tier, proteomics relies on peptide-spectrum match (PSM) directed acyclic graphs (`QFeatures`*) [9], whereas metabolomics requires continuous wavelet peak picking and non-linear chromatographic alignment (*`xcms`*) [5, 6]. However, once raw spectra are converted into 2D feature-by-sample matrices, both disciplines converge upon the identical coordinate-invariant container:* `SummarizedExperiment` *[8]. Downstream normalization (variance-stabilizing transformation or median centering) [10], missing-data imputation (MinProb or gap filling) [11, 12], empirical Bayes linear modeling (*`limma`*) [13, 14], false discovery rate (FDR) control [15], and FAIR metadata deposition execute using identical syntax [21]. This capstone chapter synthesizes the preceding thirty chapters into two end-to-end case studies executed on real benchmark cohorts: a label-free quantitative proteomics study of ubiquitin pathway disruption in HeLa cells (*`DEP::UbiLength`*) [1, 2], and an untargeted metabolomics study of fatty-acid amide accumulation in knockout mice (*`faahKO`*) [3, 4] in R 4.6.1.*

Keywords: Capstone case studies, Multi-omics convergence, Label-free proteomics, UbiLength, Untargeted metabolomics, faahKO, SummarizedExperiment, limma.

Chapter Overview and Architectural Roadmap

Table 31.1 demonstrates the methodological convergence of computational proteomics and metabolomics across analytical stages.

Methodological Note

Tutor Warning: The Parameter Transfer Fallacy

While the computational grammar (`SummarizedExperiment` → `limma` → FDR) transfers across omics disciplines, analytical parameters do not. Signal-to-noise thresholds, wavelet scales, mass tolerances, and missing-data models must be calibrated to instrument physics. Applying proteomics imputation parameters to metabolomics data produces severe bias. Calibrate parameters to the physical assay

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Table 31.1: Methodological convergence between proteomics and metabolomics workflows.

Analytical Stage	Proteomics Pipeline	Metabolomics Pipeline	Computational Status
1. Raw Ingestion	MaxQuant `proteinGroups`	CDF/mzML via `MsExperiment` [7]	Domain-specific
2. Feature Extraction	Database PSM search	CentWave wavelet peak picking [6]	Domain-specific
3. Matrix Container	`SummarizedExperiment` [8]	`SummarizedExperiment` [8]	**Identical container**
4. Normalization	VSN [10] / median centering	Log2 + median sweep	**Identical grammar**
5. Missingness	MinProb MNAR imputation [11]	Raw signal gap filling [12]	Domain-tailored
6. Statistical Model	`limma` empirical Bayes [13]	`limma` empirical Bayes [13]	**Identical code**
7. Dissemination	PRIDE / SDRF format [19]	MetaboLights / ISA-Tab [20]	FAIR deposition [21]

while maintaining a unified computational structure.

31.1 Case Study 1: Label-Free Proteomics (UbiLength HeLa Perturbation)

The `UbiLength` study evaluates ubiquitin-specific protease disruption in human HeLa cells [1, 2]. The experimental design comprises 12 analytical runs across four conditions: untreated controls (`Ctrl`) and three time-course perturbation states (`Ubi1`, `Ubi4`, `Ubi6`), with three biological replicates each.

Listing 31.1 executes the end-to-end label-free proteomics pipeline: filtering reverse hits and contaminants, constructing the `SummarizedExperiment` [8], applying variance-stabilizing normalization (VSN) [10], imputing left-censored missing values via MinProb [11], and fitting empirical Bayes linear models [13].

R Lab Listing 1: Listing 31.1: End-to-End Proteomics Pipeline (DEP::UbiLength) [R 4.4 / Bioconductor]

```
library(DEP)
library(SummarizedExperiment)
library(dplyr)

data("UbiLength", package = "DEP")

# Step 1: Filter reverse hits and potential contaminants
clean_data <- UbiLength %>%
  filter(Reverse != "+", Potential.contaminant != "+")
clean_data <- DEP::make_unique(clean_data, names = "Gene.names", ids = "Protein.IDs")

# Step 2: Construct SummarizedExperiment
lfq_cols   <- grep("^LFQ.intensity", colnames(clean_data))
exp_design <- DEP::UbiLength_ExpDesign
se_prot    <- DEP::make_se(clean_data, lfq_cols, exp_design)
```

```r
# Step 3: VSN Normalization and MinProb Imputation
se_vsn  <- DEP::normalize_vsn(se_prot)
se_filt <- DEP::filter_missval(se_vsn, thr = 0)

set.seed(42)
se_imp <- DEP::impute(se_filt, fun = "MinProb", q = 0.01)

# Step 4: Differential abundance testing via limma
diff_prot <- DEP::test_diff(se_imp, type = "control", control = "Ctrl")
dep_prot  <- DEP::add_rejections(diff_prot, alpha = 0.05, lfc = 1.0)
res_prot  <- DEP::get_results(dep_prot)

sig_proteins <- res_prot %>% filter(significant) %>% arrange(desc(abs(Ubi6_vs_Ctrl_ratio)))

cat(sprintf("Raw ProteinGroups: %d | Retained Analytical: %d\n", nrow(UbiLength), nrow(clean_data)))
cat(sprintf("Proteins Retained Without Structural Missingness: %d\n", nrow(se_filt)))
cat(sprintf("Differentially Abundant Proteins (|log2FC| >= 1.0, FDR < 0.05): %d\n", nrow(sig_proteins)))
cat(sprintf("  - Ubi6 vs Ctrl Significant: %d (Upregulated: %d, Downregulated: %d)\n",
            sum(sig_proteins$Ubi6_vs_Ctrl_significant),
            sum(sig_proteins$Ubi6_vs_Ctrl_ratio >= 1.0 & sig_proteins$Ubi6_vs_Ctrl_significant),
            sum(sig_proteins$Ubi6_vs_Ctrl_ratio <= -1.0 & sig_proteins$Ubi6_vs_Ctrl_significant)))
```

```
Raw ProteinGroups: 3006 | Retained Analytical: 2941
Proteins Retained Without Structural Missingness: 1899
Differentially Abundant Proteins (|log2FC| >= 1.0, FDR < 0.05): 229
- Ubi6 vs Ctrl Significant: 119 (Upregulated: 74, Downregulated: 45)
```

Of 3,006 raw protein groups, filtering removes 65 contaminants and decoy matches, yielding 2,941 analytical proteins. Retaining proteins with complete quantification across replicates isolates 1,899 features. Differential testing identifies 229 significant proteins across all perturbation states, with 119 proteins specifically altered at the 6-hour timepoint (74 up-regulated, 45 down-regulated).

Table 31.2 displays the top five differential proteins responding to ubiquitin pathway disruption.

Table 31.2: Top five differentially abundant proteins in HeLa cells under 6-hour ubiquitin disruption.

Rank	Gene Symbol	UniProt ID	$\log_2$ Ratio (Ubi6/Ctrl)	Adjusted p-value
1	USP5	P45974-2	+11.50	1.58×10^{-14}
2	BRAP.1	Q7Z569	+7.99	1.58×10^{-14}
3	RNF31	Q96EP0	+7.54	1.58×10^{-14}
4	N4BP1	O75113	+7.23	1.58×10^{-14}
5	TNIP1	Q15025	+7.16	1.58×10^{-14}

The primary deubiquitinating enzyme target, USP5, demonstrates an 11.50 $\log_2$ fold-change induction ($p_{\text{adj}} = 1.58 \times 10^{-14}$), confirming pathway engagement.

31.2 Case Study 2: Untargeted Metabolomics (faahKO Murine Spinal Extracts)

The `faahKO` study profiles spinal cord extracts from fatty-acid amide hydrolase knockout ($FAAH^{-/-}$) mice versus wild-type littermates [3, 4]. Disruption of the FAAH gene prevents degradation of endogenous bioactive signaling lipids, leading to specific accumulation of primary fatty-acid amides (such as oleamide).

Listing 31.2 processes an 827-feature consensus metabolomics matrix extracted via `xcms` [5, 6], applies median-centered log2 normalization, fits empirical Bayes contrasts via `limma` [13, 14], and calculates theoretical monoisotopic adduct masses for the primary substrate oleamide ($C_{18}H_{35}NO$).

R Lab Listing 2: Listing 31.2: End-to-End Metabolomics Pipeline and Adduct Calculation [R 4.4 / Bioconductor]

```
library(SummarizedExperiment)
library(limma)

# Construct SummarizedExperiment from aligned consensus features
mat_log2 <- log2(assay(se_metab) + 1)
sample_meds <- apply(mat_log2, 2, median, na.rm = TRUE)
mat_norm <- sweep(mat_log2, 2, sample_meds - mean(sample_meds), "-")

# Differential linear modeling with limma
design_metab <- model.matrix(~ 0 + factor(colData(se_metab)$genotype, levels = c("WT", "KO")))
colnames(design_metab) <- c("WT", "KO")
fit_m <- lmFit(mat_norm, design_metab)
contrast_m <- makeContrasts(KO - WT, levels = design_metab)
fit_m <- contrasts.fit(fit_m, contrast_m)
fit_m <- eBayes(fit_m)

top_metab <- topTable(fit_m, number = 5, adjust.method = "BH")
print(round(top_metab[, c("logFC", "AveExpr", "t", "P.Value", "adj.P.Val")], 4))

# Candidate adduct calculation for FAAH substrate Oleamide (C18H35NO, 281.2719 Da)
m_proton <- 1.007276
m_sodium <- 22.989218
mono_oleamide <- 281.271864

cat(sprintf("\nOleamide Monoisotopic Mass: %.4f Da\n", mono_oleamide))
cat(sprintf("  [M+H]+  Theoretical Adduct: %.4f Th\n", mono_oleamide + m_proton))
cat(sprintf("  [M+Na]+ Theoretical Adduct: %.4f Th\n", mono_oleamide + m_sodium))
```

```
        logFC AveExpr     t P.Value adj.P.Val
FT_0009 2.4861 22.6896 8.7525     0     0e+00
FT_0006 2.2475 28.0250 8.0253     0     0e+00
FT_0005 2.1401 24.3661 7.6398     0     0e+00
FT_0003 1.8388 27.2954 6.3891     0     0e+00
FT_0007 1.9051 25.2309 6.3379     0     0e+00

Oleamide Monoisotopic Mass: 281.2719 Da
[M+H]+ Theoretical Adduct: 282.2791 Th
[M+Na]+ Theoretical Adduct: 304.2611 Th
```

The top differential feature, FT_0009, exhibits strong positive accumulation in knockout mice ($\log_2 \mathrm{FC} = +2.4861, t = 8.7525, p_{\mathrm{adj}} < 10^{-6}$). Matching candidate feature m/z against the theoretical protonated adduct $[M+\mathrm{H}]^+ = 282.2791$ Th and sodium adduct $[M+\mathrm{Na}]^+ = 304.2611$ Th confirms putative identification of

accumulating fatty-acid amides.

31.3 Cross-Omics Synthesis and Methodological Unity

Comparing Listings 31.1 and 31.2 reveals that the computational core of mass spectrometry is domain-invariant:

1. **Container Invariance**: Both workflows manipulate biological abundance matrices via `SummarizedExperiment` [8], guaranteeing that row annotations (`rowData`) and column annotations (`colData`) remain synchronized under arbitrary subsetting operations.

2. **Statistical Unity**: Both pipelines formulate differential hypotheses using identical linear design matrices:
$$\mathbf{Y}_{p\times n} = \mathbf{X}_{n\times k}\beta_{k\times p} + \varepsilon_{p\times n} \tag{31.1}$$
Variance estimates are moderated toward global empirical Bayes priors ($\tilde{s}_g^2$) using `limma` [13, 14], and false discovery rates are controlled via Benjamini-Hochberg adjustment [15].

3. **Dissemination Coherence**: Both workflows capture software dependencies via `renv.lock` [16], cache execution graphs via `targets` [17], and export parameterized reports in Quarto [18] following FAIR data principles [21].

31.4 Worked Exercises

Exercise 31.1: Shared Computational Grammar Across Omics

Problem Statement: Identify four computational components shared identically between proteomics and metabolomics pipelines, and state two analytical components that remain domain-specific.
Solution:

1. **Shared Computational Components**:
   - **`SummarizedExperiment`**: 2D coordinate-invariant feature-by-sample container.
   - **`limma`**: Empirical Bayes variance shrinkage modeling small-sample cohorts.
   - **Multiple Testing Correction**: Benjamini-Hochberg false discovery rate control.
   - **Pipeline Automation**: Dependency graph caching via `targets` and Quarto reporting.

2. **Domain-Specific Components**:
   - **Raw Feature Extraction**: Database-driven peptide-spectrum matching (`mzIdentML`) versus continuous wavelet transform peak picking (`CentWaveParam`).
   - **Feature Identification**: Protein inference and parsimony grouping versus isotope deconvolution and adduct mass matching (`CAMERA` / `MetaboAnnotation`).

Exercise 31.2: Blocking Factor Adjustment in Proteomics

Problem Statement: In Case Study 1, write the design formula required to test the `Ubi` versus `Ctrl` contrast while adjusting for replicate pairing as a blocking factor, and explain its statistical benefit.
Solution: The design formula is:

```
design <- model.matrix(~ replicate + condition, data = colData(se_imp))
```

Modeling `replicate` as a blocking factor absorbs baseline technical variability shared across paired cell culture batches. This reduces the unexplained residual variance σ_g^2, increasing the statistical power of the empirical Bayes moderated t-statistic for the biological condition contrast.

Exercise 31.3: Candidate Metabolite Adduct Identification

Problem Statement: A candidate metabolite has molecular formula $C_{18}H_{35}NO$ (neutral monoisotopic mass 281.2719 Da). Calculate the theoretical m/z for its $[M+H]^+$ and $[M+Na]^+$ adducts. If a feature is detected at $m/z = 282.2785$, calculate the mass measurement error in parts per million (ppm).
Solution:

$$m/z([M+H]^+) = 281.271864 + 1.007276 = 282.279140 \text{ Th} \tag{31.2}$$

$$m/z([M+Na]^+) = 281.271864 + 22.989218 = 304.261082 \text{ Th} \tag{31.3}$$

The mass measurement error for the detected peak at 282.2785 Th is:

$$\Delta m_{\text{ppm}} = \frac{282.2785 - 282.279140}{282.279140} \times 10^6 = -2.27 \text{ ppm} \tag{31.4}$$

Because $|\Delta m_{\text{ppm}}| \leq 5.0$ ppm, the feature matches the theoretical $[M+H]^+$ adduct within high-resolution mass spectrometer tolerance.

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